WEEK 1 · LECTURE L2 · UNIT I

Why HR Is Going Digital

Industry 4.0, artificial intelligence and hybrid work, and what the three of them together are doing to the HR function and to the nature of work itself.

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DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Explain the three forces reshaping HR (Industry 4.0, artificial intelligence, hybrid and platform work) and why HR cannot opt out of them.
- Distinguish digitization, digitalization and transformation, and classify real HR examples correctly.
- Apply Zuboff, Autor, Levy and Murnane, Orlikowski and Verhoef et al. to an HR change, and state the limits of each.
- Argue why imported digital HR frameworks need adapting to Indian organizational conditions.

COURSE OUTCOMES

CLO 1 (concepts and drivers of digital transformation in HR) and CLO 5 (ethics and strategic alignment).

OPENING QUESTION

What actually changes when HR goes digital?

Who gets faster

And who quietly gets slower, because the new system was not designed around their device, language or job?

What judgment is encoded

Which decision moves from a person to a rule, and who can still see that rule and question it?

Whose data moves

What does the organization now know about each employee that it did not know before, and who controls it?

Hold these three questions open. Every theory in this lecture is an answer to one of them.

THE DRIVERS

Three forces, arriving together

I. Industry 4.0

Cyber-physical systems, the Internet of Things, cloud and connectivity collapse the distance between the shop floor and the data layer. Work becomes measurable in real time.

II. Artificial intelligence

Prediction becomes cheap. Screening, matching, ranking and forecasting move from human judgment to model output: fast and scalable, but harder to explain.

III. Hybrid and platform work

Where work happens, and under what contract, both come loose from the office and the payroll. The workforce stops being one population.

None of these is an HR initiative, yet all three land on HR. That is the core argument for why the HR function must change shape.

SCHWAB (2016)

Force I: the fourth industrial revolution

Revolution	Technology	What it meant for people management
First (from c. 1784)	Steam and mechanization	HR did not exist; labor was hired at the factory gate.
Second (from c. 1870)	Electricity and mass production	Personnel administration appears: records, payroll, discipline.
Third (from c. 1969)	Electronics and information technology	HRM professionalizes; systems of record (the HRIS) arrive.
Fourth (now)	Cyber-physical systems, AI, data	Every process emits data; HR is asked to decide with it, and to answer for it.

EXAM LENS

Schwab argues the fourth revolution differs in velocity, breadth and depth, and systems impact. A strong answer uses all three criteria, not just "it is faster".

SCULLY-RUSS AND TORRACO (2020)

What technology does to work itself

Tasks split from jobs

A job becomes a shifting portfolio of tasks, some of which move to machines each year.

Skills develop a half-life

Technical skills lose value faster, so learning becomes continuous infrastructure, not a one-time event.

Work emits data

Logins, scans, routes and clicks are recorded whether or not anyone designed a measurement system.

The firm loses its edges

Employees, contractors, apprentices and platform workers work side by side under different contracts.

EXAM LENS

Each shift lands on an HR process first: job design, learning, privacy and workforce planning. Link each shift to the HR process it disrupts.

FORCE II

Where algorithms already sit in the employee lifecycle

Stage	Typical algorithmic use	What improves	What can get worse
Attract and screen	CV ranking, keyword filters, chatbot pre-screening	Speed and consistency at volume	Past bias is learned and scaled
Onboard and learn	Document verification, content recommendation	Administrative load falls	Reasons for decisions become invisible
Perform and reward	Goal cascades, calibration, pay modeling	Patterns no manager could see unaided	Accountability diffuses: "the model decided"
Predict and retain	Flight risk scores, succession, headcount forecasts	Earlier intervention	Prediction treated as fact

GRATTON (2021)

Force III: hybrid work, and the equity trap

Jobs and tasks

Which tasks need co-location, concentration, coordination or cooperation?

Employee preferences

Autonomy over place and time is now a retention variable, not a perk.

Projects and workflows

Redesign the workflow, or hybrid simply degrades it.

Inclusion and fairness

Who is eligible for flexibility, and who is structurally excluded?

The trap: a store associate, a factory operator or a delivery partner cannot work from home. If flexibility is designed only for laptop users, hybrid work divides the workforce in two, and HR built the wall.

CONTEXTUALIZING THE FORCES

The Indian layer: why imported frameworks slip

The divide is inside the firm

Corporate offices are often well digitized while store floors, factories and warehouses remain manual. Vernacular languages, shared devices and thin bandwidth are common.

Platform work is scaling fast

NITI Aayog (2022) estimated about 7.7 million gig workers in 2020 to 21, projected to reach about 23.5 million by 2029 to 30.

The law has arrived

The Digital Personal Data Protection Act, 2023 governs employee data and consent; the Code on Social Security, 2020 and state laws begin to reach platform work.

Public digital rails

India Stack (Aadhaar based verification, DigiLocker, UPI) enables onboarding and payment designs most global HR vendors never built for, with privacy trade-offs of their own.

THE CONSEQUENCE

Where the three forces meet: four shifts in the HR role

From records to evidence

HR must prove effects, not report activity. "We ran 40 sessions" stops being an answer.

From process owner to product owner

HR services become interfaces people either use or abandon; adoption becomes an HR metric.

From administrator to governor

Someone must own algorithmic fairness, consent and data residency. In practice, that is HR.

From employee relations to workforce relations

Employees, contractors, apprentices and platform workers: several populations, one duty of care.

THE DISTINCTION THE COURSE TURNS ON

Digitization is not transformation

Digitization: the same process, on a screen	Transformation: a different process, because the constraint moved
The form becomes a PDF	The step is questioned: why does it exist at all?
The register becomes a spreadsheet	The decision right moves to where the data is
The queue becomes an inbox	The service reaches people it never reached
The old rule runs automatically, including its errors	Roles are redesigned, not merely re-tooled
Cost falls a little; experience rarely improves	The measure of success itself changes

EXAM LENS

Many "digital HR" programs are digitization wearing a transformation budget. Examiners reward answers that show which constraint moved, not answers that list technologies.

APPLY THE DISTINCTION

Classify each example: digitization or transformation?

1

An annual paper engagement survey is reissued as a PDF form emailed to all staff.

2

Leave applications move from a register to a shared spreadsheet that HR updates each evening.

3

Frontline staff get a vernacular mobile app that shows leave balances and approves leave instantly.

4

An applicant tracking system ranks CVs using the same rule hiring managers always used.

5

A chatbot answers payroll queries, and payroll clerks are retrained as data quality owners.

Your task

Mark each D or T, then justify the hardest one using Verhoef et al.'s three stages.

THEORY IN DEPTH · ZUBOFF (1988)

Technology can automate, or it can informate

THE CORE IDEA

Automating replaces human effort. Informating produces new information about the work itself.

HOW IT WORKS

The same system does both. The choice is whether the new information empowers workers or only watches them.

LIMITS AND CRITIQUE

Written before the internet; Zuboff (2019) later argued that informating turned into surveillance for profit.

IN PRACTICE

An employee self service app automates leave requests; it also informates, because every login tells HR who uses it and who does not.

EXAM LENS

Use Zuboff to ask who benefits from the data a system creates. Pair her 1988 and 2019 books to show how the argument evolved.

THEORY IN DEPTH · AUTOR, LEVY AND MURNANE (2003)

Computers replace routine tasks, not whole jobs

THE CORE IDEA

Jobs are bundles of tasks. Machines substitute for routine tasks and complement non-routine ones.

HOW IT WORKS

Routine cognitive tasks (record keeping) and routine manual tasks go first; judgment and personal service resist automation.

LIMITS AND CRITIQUE

AI now reaches some non-routine cognitive tasks, so the routine line keeps moving.

IN PRACTICE

An HR officer's data entry is routine; their process knowledge and judgment are not. Redesign the job rather than delete it.

EXAM LENS

Strong answers analyze tasks, not job titles. Note how generative AI challenges the original routine and non-routine boundary.

THEORY IN DEPTH · ORLIKOWSKI (1992)

Technology shapes people, and people reshape technology

THE CORE IDEA

The duality of technology: systems are designed by people, then constrain them, and are reinterpreted in use.

HOW IT WORKS

Users adapt, work around, or ignore features. The system that runs is not the system that was bought.

LIMITS AND CRITIQUE

The idea explains a great deal after the fact but predicts little in advance.

IN PRACTICE

Employees often keep private spreadsheets alongside an official HRIS, a workaround showing that the official system did not fit their work.

EXAM LENS

Use Orlikowski to explain why adoption figures mislead: high usage of a workaround can signal failure of the official system.

THEORY IN DEPTH · VERHOEF ET AL. (2021)

Three stages: digitization, digitalization, transformation

THE CORE IDEA

Digitization turns analog into digital. Digitalization changes processes. Transformation changes the business model.

HOW IT WORKS

Each stage needs different resources: tools first, then process redesign, then new capabilities and structures.

LIMITS AND CRITIQUE

Real organizations sit in different stages in different functions at once; the stages are a lens, not a ladder.

IN PRACTICE

A PDF survey is digitization; a vernacular mobile leave flow is digitalization; HR acting as a data-led strategic partner is transformation.

EXAM LENS

The three-stage vocabulary sharpens the digitization versus transformation distinction. Define all three precisely before applying them.

COMPARE AND EVALUATE

Four lenses on the same digital change

Theory	Question it answers	Best used to	Main weakness
Zuboff (1988; 2019)	Who gains from the data?	Analyze power, surveillance and empowerment	Can underplay benefits to workers
Autor, Levy and Murnane (2003)	Which tasks will change?	Redesign jobs and plan reskilling	The routine line keeps moving
Orlikowski (1992)	Why does the system in use differ from the one bought?	Explain adoption and workarounds	Weak at prediction
Verhoef et al. (2021)	How deep is the change?	Classify and sequence initiatives	Stages blur in practice

EXAM LENS

Questions that ask you to "critically evaluate" reward choosing the right lens for the question and naming its weakness.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"Most digital HR programs are digitization projects wearing a transformation budget." Critically evaluate this statement with reference to Verhoef et al. (2021) and Indian organizational examples. **20 marks**

A strong answer: defines digitization, digitalization and transformation precisely; tests each example against which constraint moved; uses Orlikowski or Zuboff to explain why programs stall; concludes with conditions under which the statement is false.

Industry 4.0, artificial intelligence and hybrid work are external forces, not HR initiatives. Discuss their combined implications for the HR function in India. **20 marks**

A strong answer: applies Schwab's three criteria; links each force to specific HR processes; addresses the frontline digital divide, the DPDP Act and platform work; evaluates the equity trap rather than only describing it.

Using Autor, Levy and Murnane (2003), explain why automation changes tasks rather than eliminating jobs. What is the main limitation of this view today? **10 marks**

A strong answer: explains the task model with an HR example, then names the moving routine line under AI as the limitation.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Three external forces, not three HR projects**
   Industry 4.0, AI and hybrid work arrived together and landed on HR; it cannot opt out of any of them.

2. **Digitization moves a process onto a screen; transformation asks why it exists**
   Show which constraint moved, and use Verhoef et al.'s three stages to classify precisely.

3. **Reach is the constraint, not features**
   A design that works only for the desk-based, English-fluent minority divides the workforce rather than transforming it.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Schwab, K. (2016). *The Fourth Industrial Revolution*, Introduction and Part I.
- Stone, D. L., Deadrick, D. L., Lukaszewski, K. M., & Johnson, R. (2015). The influence of technology on the future of human resource management. *Human Resource Management Review, 25*(2), 216–231.
- Next week: Westerman, Bonnet and McAfee (2014), Chapters 1 and 2; Scully-Russ and Torraco (2020), pp. 73–77.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics, 118*(4), 1279–1333. https://doi.org/10.1162/003355303322552801

Gratton, L. (2021). How to do hybrid right. *Harvard Business Review, 99*(3).

NITI Aayog. (2022). *India's booming gig and platform economy: Perspectives and recommendations on the future of work*. Government of India.

Orlikowski, W. J. (1992). The duality of technology: Rethinking the concept of technology in organizations. *Organization Science, 3*(3), 398–427. https://doi.org/10.1287/orsc.3.3.398

Schwab, K. (2016). *The fourth industrial revolution*. World Economic Forum.

Scully-Russ, E., & Torraco, R. J. (2020). The changing nature and organization of work: An integrative review of the literature. *Human Resource Development Review, 19*(1), 66–93. https://doi.org/10.1177/1534484319886394

Sources for this lecture (continued)

Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research, 122*, 889–901. https://doi.org/10.1016/j.jbusres.2019.09.022

Zuboff, S. (1988). *In the age of the smart machine: The future of work and power*. Basic Books.

Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.